

Problem Set 1

Instructions

- Submit individually through the course page on **Campus Virtual**.
- Upload a single PDF with your answers to the analytical parts, together with your code for the numerical parts and the requested figures, if applicable. The preferred language is MATLAB.

Exercise 1: Exact aggregation and the dynamics of inequality

Consider an infinite-horizon production economy without uncertainty, populated by I consumers indexed by i . Each consumer is born with a capital endowment k_0^i , supplies one unit of labor inelastically, and has preferences

$$\sum_{t=0}^{\infty} \beta^t u(c_t^i), \quad u(c) = \frac{(c - \bar{c})^{1-\sigma} - 1}{1-\sigma}, \quad \bar{c} \geq 0,$$

where $\bar{c}$ is a subsistence level; when $\sigma = 1$, $u(c) = \ln(c - \bar{c})$. Output is produced with the Cobb–Douglas technology $Y_t = K_t^\alpha L_t^{1-\alpha}$, where K_t is aggregate capital and $L_t = I$; capital depreciates at rate δ . Let C_t denote aggregate consumption, $k_t \equiv K_t/I$ and $c_t \equiv C_t/I$ capital and consumption per consumer, $R_t = 1 - \delta + \alpha k_t^{\alpha-1}$ the gross return on capital and $w_t = (1 - \alpha)k_t^\alpha$ the wage. Markets are Arrow–Debreu (complete). Let p_t denote the date-0 price of date- t consumption (with $p_0 = 1$), and W_0^i the value of consumer i 's endowments (initial capital plus the present value of labor income).

- (a) Suppose first $\bar{c} = 0$. Show that these preferences are homothetic and that the economy admits *exact aggregation*: equilibrium prices and aggregates coincide with those of an economy with a single consumer who owns all the wealth.
- (b) Now suppose $\bar{c} > 0$. Show that individual equilibrium consumption can be written as

$$c_t^i = \bar{c} + b_t(p) \left(W_0^i - \bar{c} \sum_{\tau} p_{\tau} \right),$$

for some function $b_t(p)$ that does not depend on i (assume $W_0^i > \bar{c} \sum_{\tau} p_{\tau}$ for all i). Show that aggregate consumption depends only on aggregate wealth, so a representative consumer still exists. *What is lost relative to part (a)? Compare each consumer's consumption share with their wealth share.*

- (c) **Total wealth.** The Arrow–Debreu budget constraint can be written sequentially as $c_t^i + k_{t+1}^i = R_t k_t^i + w_t$, with $p_{t+1}/p_t = 1/R_{t+1}$. Define consumer i 's total wealth at the beginning

of period t as $W_t^i = \sum_{\tau \geq t} (p_\tau/p_t) c_\tau^i$, and let $A_t = \sum_{\tau \geq t} p_\tau/p_t$.

- (i) Show that $W_t^i = R_t k_t^i + h_t$, where h_t is the present value of wages from t on: total wealth is financial plus human wealth.
- (ii) Show that total wealth net of the cost of subsistence, $W_t^i - \bar{c} A_t$, grows at a rate common to all consumers.
- (iii) Show that

$$\frac{W_{t+1}^i}{W_t^i} = \frac{p_t}{p_{t+1}} \left(1 - m_t(p) - \frac{\Theta_t(p, \bar{c})}{W_t^i} \right),$$

where $m_t(p)$ is the marginal propensity to consume out of *date- t* wealth, common to all consumers, and $\Theta_t(p, \bar{c})$ is a function you should characterize explicitly. How does $m_t(p)$ relate to $b_t(p)$ in part (b)?

- (d) Show that in steady state $\Theta_t(p, \bar{c}) = 0$, but that along the transition to the steady state (with $k_t < k^*$) we have $\Theta_t(p, \bar{c}) > 0$, for any $\sigma > 0$.

- (e) **Three cross sections.** Let W_t denote average total wealth, and consider relative consumption c_t^i/c_t , relative total wealth W_t^i/W_t and relative financial wealth k_t^i/k_t . Say that a cross section *diverges (converges)* if $|x_t^i/x_t - 1|$ rises (falls) for every consumer with $x_t^i \neq x_t$. Along the transition from $k_0 < k^*$:

- (i) Show that relative total wealth is constant if $\bar{c} = 0$ and diverges if $\bar{c} > 0$.
- (ii) Show that $c_t^i - \bar{c}$ grows at a rate common to all consumers. Conclude that relative consumption is constant if $\bar{c} = 0$ and diverges if $\bar{c} > 0$.
- (iii) Show that $k_{t+1}^i - k_{t+1} = R_t(1 - m_t)(k_t^i - k_t)$, and that relative financial wealth converges if and only if $w_t - m_t h_t > \Theta_t$. Interpret both sides of this inequality. How can financial wealth converge while total wealth diverges?

- (f) **Numerical.** Calibrate $\{\beta, \sigma, \alpha, \delta\}$ to reasonable values and simulate the transition from $k_0 = k^*/2$ for three values of $\bar{c}$: zero, a small one and a large one (as fractions of steady-state consumption). For two consumers with different initial capital, plot relative consumption, relative total wealth and relative financial wealth. Interpret: *along the transition, does inequality rise or fall? Does the answer depend on which cross section you look at, and on $\bar{c}$? What does this say about the restrictions that the existence of a representative consumer places on the dynamics of the distribution?*

Exercise 2: Skill dispersion and “wedges”

Consider an economy populated by I households, indexed by $i = 1, \dots, I$, with I even. Each period, household i draws a skill level $\theta_t^i > 0$: one hour worked by household i supplies θ_t^i efficiency units of labor. Skills come from a two-point distribution: each period, half of the households draw $\theta_t^i = 1 + \varepsilon_t$ and the other half draw $\theta_t^i = 1 - \varepsilon_t$, with $0 \leq \varepsilon_t < 1$. Which households draw the high skill is random and independent over time, so every household faces idiosyncratic skill risk, while average skill is $\frac{1}{I} \sum_i \theta_t^i = 1$ in every period. The dispersion of skills, ε_t , may vary over time. Households have identical preferences

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{c_t^{1-\sigma}}{1-\sigma} - \psi \frac{n_t^{1+1/\eta}}{1+1/\eta} \right],$$

where c is consumption, n is hours worked, $\psi > 0$, and $\eta > 0$ is the Frisch elasticity of labor supply (with $\ln c$ in place of $c^{1-\sigma}/(1-\sigma)$ when $\sigma = 1$). All aggregates are per household. Output is produced with capital and efficiency units of labor,

$$Y_t = K_t^\alpha H_t^{1-\alpha}, \quad H_t = \frac{1}{I} \sum_i \theta_t^i n_t^i,$$

and $K_{t+1} = (1 - \delta)K_t + Y_t - C_t$, with $C_t = \frac{1}{I} \sum_i c_t^i$. Markets are complete, so the equilibrium allocation solves a planner's problem with Pareto weights $a^i > 0$ that are constant over time.

(a) The distribution problem. Take any vector of skills $(\theta_t^1, \dots, \theta_t^I)$. Given aggregates (C_t, H_t) , the planner allocates consumption and hours across households to maximize

$$\frac{1}{I} \sum_i a^i \left[\frac{(c^i)^{1-\sigma}}{1-\sigma} - \psi \frac{(n^i)^{1+1/\eta}}{1+1/\eta} \right] \quad \text{subject to} \quad \frac{1}{I} \sum_i c^i = C_t, \quad \frac{1}{I} \sum_i \theta_t^i n^i = H_t.$$

Show that the value of this problem is

$$\Gamma \frac{C_t^{1-\sigma}}{1-\sigma} - \Theta_t \psi \frac{H_t^{1+1/\eta}}{1+1/\eta},$$

where

$$\Gamma = \left(\frac{1}{I} \sum_i (a^i)^{1/\sigma} \right)^\sigma, \quad \Theta_t = \left(\frac{1}{I} \sum_i (a^i)^{-\eta} (\theta_t^i)^{1+\eta} \right)^{-1/\eta}.$$

Conclude that aggregates behave as if chosen by a representative consumer with a preference shock Θ_t on efficiency units of labor. Which households work more?

(b) Equal weights. Let $a^i = 1$ for all i , and define $M_k(\varepsilon) \equiv \frac{1}{2}[(1 + \varepsilon)^k + (1 - \varepsilon)^k]$. Show that

$\Theta_t = M_{1+\eta}(\varepsilon_t)^{-1/\eta}$, and that physical hours $N_t = \frac{1}{I} \sum_i n_t^i$ satisfy

$$N_t = \frac{M_\eta(\varepsilon_t)}{M_{1+\eta}(\varepsilon_t)} H_t.$$

Show that $N_t \leq H_t$, with equality only if $\varepsilon_t = 0$ (hint: θ^η is increasing in θ), and that for $\eta = 1$ both Θ_t and N_t/H_t equal $1/(1 + \varepsilon_t^2)$. Explain the intuition. Rewrite the technology in terms of physical hours and show that dispersion acts like a labor-augmenting technology shock. Give an example of an economic event that would raise ε_t .

(c) Through the lens of a representative-agent model. An outside observer sees Y_t, C_t, K_t and physical hours N_t , but neither θ_t^i nor ε_t . The observer fits a representative-agent model with preferences $C^{1-\sigma}/(1-\sigma) - \psi N^{1+1/\eta}/(1+1/\eta)$ and technology $Y = A_t K^\alpha N^{1-\alpha}$, and measures three “wedges”:

- an efficiency wedge $A_t = Y_t/(K_t^\alpha N_t^{1-\alpha})$;
- a labor wedge $\tau_{n,t}$, defined by $(1 - \tau_{n,t})(1 - \alpha) \frac{Y_t}{N_t} = \psi N_t^{1/\eta} C_t^\sigma$;
- an investment wedge $\tau_{x,t}$, defined by

$$(1 + \tau_{x,t}) C_t^{-\sigma} = \beta E_t \left[C_{t+1}^{-\sigma} \left(\alpha \frac{Y_{t+1}}{K_{t+1}} + (1 - \delta)(1 + \tau_{x,t+1}) \right) \right].$$

With equal weights, show that

$$A_t = \left(\frac{M_{1+\eta}(\varepsilon_t)}{M_\eta(\varepsilon_t)} \right)^{1-\alpha}, \quad 1 - \tau_{n,t} = \frac{M_\eta(\varepsilon_t)^{1+1/\eta}}{M_{1+\eta}(\varepsilon_t)}, \quad \tau_{x,t} = 0.$$

Simplify A_t and $\tau_{n,t}$ for $\eta = 1$. For $\alpha = 1/3$ and $\eta = 1$, plot A_t and $\tau_{n,t}$ for $\varepsilon_t \in [0, 0.3]$.

(d) Business cycle accounting (Chari, Kehoe and McGrattan, 2007) uses measured wedges to guide researchers toward the margins where frictions matter. A tempting reading is that each wedge stands in for a friction in the corresponding market. In this economy there are no frictions, no policy, and average skill never changes. What would a business-cycle accountant conclude from a rise in ε_t ? Which wedges does skill dispersion contaminate, which does it leave untouched, and why? Is a wedge evidence of *where* a friction lives — or of whether there is a friction at all?

Exercise 3: How do you test whether markets are complete?

In this exercise you will test the complete-markets hypothesis on data generated by a model in which, by construction, markets *are* complete — first with heterogeneous risk aversion, then with identical risk aversion.

Consider a pure-exchange, Arrow–Debreu economy with two types of households, $i \in \{1, 2\}$, each of mass $1/2$. Aggregate income per household $Y_t \in \{Y_L, Y_H\}$ is i.i.d. over time, with $\Pr(Y_H) = q$. Let $z_t = +1$ if $Y_t = Y_H$ and $z_t = -1$ otherwise. Every household of type i receives the endowment

$$y_{i,t} = (1 + \kappa_i z_t) Y_t, \quad \kappa_1 = \kappa, \quad \kappa_2 = -\kappa, \quad 0 \leq \kappa < 1,$$

so endowments add up to Y_t and, for $\kappa > 0$, type 1's income is more cyclical than type 2's. Households share a common discount factor β , and preferences are CRRA with *type-specific* risk aversion: $u_i(c) = c^{1-\gamma_i}/(1-\gamma_i)$, with $u_i(c) = \ln c$ when $\gamma_i = 1$.

- (a) Write the planner's problem with Pareto weights (a_1, a_2) and show that the efficient allocation in each state solves

$$a_1 c_1(Y)^{-\gamma_1} = a_2 c_2(Y)^{-\gamma_2}, \quad \frac{1}{2} c_1(Y) + \frac{1}{2} c_2(Y) = Y,$$

which, in general, have no closed-form solution unless $\gamma_1 = \gamma_2$.

- (b) **Numerical.** Throughout, treat the Pareto weights as given (as if initial wealth had been redistributed to support them). Set $(a_1, a_2) = (\frac{1}{2}, \frac{1}{2})$, $Y_L = 0.9$, $Y_H = 1.1$. Compute $c_i(Y)$ for $(\gamma_1, \gamma_2) = (1, 5)$ and for $(\gamma_1, \gamma_2) = (2, 2)$, and report the consumption shares $\omega_i(Y) = c_i(Y)/Y$ (relative to average income, so $\frac{1}{2}\omega_1 + \frac{1}{2}\omega_2 = 1$). Show that the shares vary with Y only when $\gamma_1 \neq \gamma_2$. Which type bears more aggregate risk, and why is that efficient?
- (c) **Numerical.** Simulate $T = 1,000$ periods with $q = \frac{1}{2}$ and 200 households of each type. Measured log consumption of household h of type i is $\ln c_{h,t}^{obs} = \ln c_i(Y_t) + \xi_{h,t}$, with $\xi_{h,t}$ i.i.d. $\mathcal{N}(0, 0.1^2)$ measurement error. Set $\kappa = 0.1$ and run the standard risk-sharing regression of consumption growth on own income growth, with *time effects*:

$$\Delta \ln c_{h,t+1}^{obs} = \alpha_{t+1} + \alpha_2 \Delta \ln y_{h,t+1} + \nu_{h,t+1},$$

where $y_{h,t} = y_{i,t}$ for every household h of type i , and α_{t+1} is one constant per period (equivalently, demean both variables period by period). Report $\hat{\alpha}_2$ for $(\gamma_1, \gamma_2) = (1, 5)$ and for $(2, 2)$. Show that $\hat{\alpha}_2 \neq 0$ in the first case even though, by construction, **markets are complete**.

- (d) Repeat (c) with $\kappa = -0.1$ and explain the sign of $\hat{\alpha}_2$. What happens when $\kappa = 0$?

- (e) Discuss: what generates the rejection of complete markets in (c)? Is it a problem of incomplete markets, or of a misspecified test? What does it imply for how we should interpret rejections of complete markets such as Cochrane's (1991)?

References

- Chari, V. V., P. J. Kehoe and E. R. McGrattan (2007). "Business Cycle Accounting." *Econometrica* 75(3): 781–836.
- Cochrane, J. H. (1991). "A Simple Test of Consumption Insurance." *Journal of Political Economy* 99(5): 957–976.